

TASK 1

```
print("welcome to my first python program")

Name = input("enter your name:")
college_name = input("enter your college name:")
branch_name = input("enter your branch name:")

print (f"Name: {Name}")
print (f"college_name:{college_name}")
print (f"branch_name:{branch_name}")
```

output task 1

C:\Users\hp\Desktop\.venv\Scripts\python.exe

"C:\Users\hp\Desktop\first program.py"

welcome to my first python program

enter your name: ayush gupta

enter your college name: thakur shivkumar engineering college

enter your branch name: computer science

Name: ayush gupta

college_name: thakur shivkumar engineering college

branch_name: computer science

Process finished with exit code 0

TASK 2

```
age = 20
percentage = 50.5
course = " computer science "
student = True

print(f"Value: {age} | Data Type: {type(age)}")
print(f"Value: {percentage} | Data Type: {type(percentge)}")
print(f"Value: {course} | Data Type: {type(course)}")
print(f"Value: {student} | Data Type: {type(student)}")
```

OUTPUT TASK 2

C:\Users\hp\Desktop\.venv\Scripts\python.exe

"C:\Users\hp\Desktop\second task.py"

Value: 20 | Data Type: <class 'int'>

Value: 50.5 | Data Type: <class 'float'>

Value: computer science | Data Type: <class 'str'>

Value: True | Data Type: <class 'bool'>

Process finished with exit code 0

TASK 3

```
a = int(input("Enter a number: "))
b = int(input("Enter another number: "))
addition = a + b
subtraction = a - b
multiplication = a * b
division = a / b
modulus = a % b
print(f"addition ({a} + {b}) ={addition}")
print(f"subtraction ({a} - {b}) ={subtraction}")
print(f"multiplication ({a} * {b}) ={multiplication}")
print(f"division ({a} / {b}) ={division}")
print(f"modulus ({a} % {b}) ={modulus}")
```

OUTPUT TASK 3

C:\Users\hp\Desktop\.venv\Scripts\python.exe

"C:\Users\hp\Desktop\third task.py"

Enter a number: 10

Enter another number: 20

addition (10 + 20) =30

subtraction (10 - 20) =-10

multiplication (10 * 20) =200

division (10 / 20) =0.5

modulus (10 % 20) =10

Process finished with exit code 0

TASK 4

```
marks = int(input("Enter your marks: "))
if marks >= 90:
    grade = "A"
elif marks >= 75:
    grade = "B"
elif marks >= 60:
    grade = "C"
else:
    grade = "Fail"
print(f"your grade is: {grade}")
```

OUTPUT TASK 4

C:\Users\hp\Desktop\.venv\Scripts\python.exe

"C:\Users\hp\Desktop\fourth task.py"

Enter your marks: 44

your grade is: Fail

Process finished with exit code 0

TASK 5

```
print("--- Numbers from 1 to 20 ---")
for i in range(1, 21):
    print(i)

print("\n--- Multiplication Table ---")
num = int(input("Enter a number for table: "))
for i in range(1, 11):
    print(f"{num} x {i} = {num * i}")

print("\n--- Even Numbers from 1 to 50 ---")
x = 1
while x <= 50:
    if x % 2 == 0:
        print(x)
    x = x + 1
```

OUTPUT TASK 5

C:\Users\hp\Desktop\.venv\Scripts\python.exe
"C:\Users\hp\Desktop\fifth task.py"

--- Numbers from 1 to 20 ---

1
2
3
4
5
6
7
8
9
10

11

12

13

14

15

16

17

18

19

20

--- Multiplication Table ---

Enter a number for table: 2

$$2 \times 1 = 2$$

$$2 \times 2 = 4$$

$$2 \times 3 = 6$$

$$2 \times 4 = 8$$

$$2 \times 5 = 10$$

$$2 \times 6 = 12$$

$$2 \times 7 = 14$$

$$2 \times 8 = 16$$

$$2 \times 9 = 18$$

$$2 \times 10 = 20$$

--- Even Numbers from 1 to 50 ---

2

4

6

8

10

12

14

16

18

20

22

24

26

28

30

32

34

36

38

40

42

44

46

48

50

Process finished with exit code 0

TASK 6

```
def calculate_square(number):  
    return number * number  
def calculate_average(n1, n2, n3):  
    return (n1 + n2 + n3) / 3  
print("=== Python Functions Program ===")  
num_for_square = int(input("Enter a number to find its square: "))  
square_result = calculate_square(num_for_square)  
print(f"Square of {num_for_square} is: {square_result}")  
  
print("\n-----")  
print("Enter three numbers to find their average:")  
num1 = float(input("First number: "))  
num2 = float(input("Second number: "))  
num3 = float(input("Third number: "))  
  
average_result = calculate_average(num1, num2, num3)  
print(f"Average of these three numbers is: {average_result}")
```

OUTPUT TASK 6

C:\Users\hp\Desktop\.venv\Scripts\python.exe

"C:\Users\hp\Desktop\six task.py"

=== Python Functions Program ===

Enter a number to find its square: 64

Square of 64 is: 4096

Enter three numbers to find their average:

First number: 3

Second number: 6

Third number: 9

Average of these three numbers is: 6.0

Process finished with exit code 0

TASK 7

```
text = "Python Programming"
print("=== 1. String Operations ===")
print("Original:", text)
print("Upper   :", text.upper())
print("Lower   :", text.lower())
print("Replace  :", text.replace("Python", "Java"))
print("Find 'Pro':", text.find("Pro"))
fruits = ["Apple", "Banana"]
print("\n=== 2. List Operations ===")
print("Original List:", fruits)
fruits.append("Orange")
print("After Append :", fruits)
fruits.remove("Banana")
print("After Remove :", fruits)
fruits.sort()
print("After Sort   :", fruits)
colors = ("Red", "Green", "Blue")
print("\n=== 3. Tuple & Indexing ===")
print("Tuple       :", colors)
print("First Color  :", colors[0])
print("Second Color :", colors[1])
student_info = {
    "name": "ayush",
    "course": "B.Tech",
    "year": "2nd Year"
}
print("\n=== 4. Dictionary ===")
print("Student Name  :", student_info["name"])
print("Student Course:", student_info["course"])
numbers_set = {10, 20, 30}
print("\n=== 5. Set Operations ===")
print("Original Set:", numbers_set)
numbers_set.add(40)
print("After Add   :", numbers_set)
numbers_set.remove(20)
print("After Remove:", numbers_set)
```

OUTPUT TASK 7

```
C:\Users\hp\Desktop\.venv\Scripts\python.exe  
"C:\Users\hp\Desktop\seven task.py"
```

=== 1. String Operations ===

Original: Python Programming

Upper : PYTHON PROGRAMMING

Lower : python programming

Replace : Java Programming

Find 'Pro': 7

=== 2. List Operations ===

Original List: ['Apple', 'Banana']

After Append : ['Apple', 'Banana', 'Orange']

After Remove : ['Apple', 'Orange']

After Sort : ['Apple', 'Orange']

=== 3. Tuple & Indexing ===

Tuple : ('Red', 'Green', 'Blue')

First Color : Red

Second Color : Green

=== 4. Dictionary ===

Student Name : ayush

Student Course: B.Tech

=== 5. Set Operations ===

Original Set: {10, 20, 30}

After Add : {40, 10, 20, 30}

After Remove: {40, 10, 30}

Process finished with exit code 0

TASK 8

```
print("=== File Handling Program ===")

with open("my_intro.txt", "w") as file:
    file.write("Hello, my name is ayush Gupta.\n")
    file.write("I am studying Computer Science Engineering.\n")
    file.write("This is a text file created using Python.")

print("\n--- Reading File Contents ---")

with open("my_intro.txt", "r") as file:
    content = file.read()
    print(content)
```

OUTPUT TASK 8

C:\Users\hp\Desktop\.venv\Scripts\python.exe

"C:\Users\hp\Desktop\task 8.py"

=== File Handling Program ===

--- Reading File Contents ---

Hello, my name is ayush Gupta.

I am studying Computer Science Engineering.

This is a text file created using Python.

Process finished with exit code 0

TASK 9

```
# छात्रों का रिकॉर्ड स्टोर करने के लिए एक लिस्ट (Data Storage)
student_records = []

# 1. छात्र जोड़ने का फंक्शन
def add_student():
    name = input("Enter student name: ")
    roll_no = input("Enter roll number: ")
    course = input("Enter course: ")

    # डिक्शनरी में जानकारी सुरक्षित करना
    student = {"name": name, "roll_no": roll_no, "course": course}
    student_records.append(student)
    print(f"✓ {name} का रिकॉर्ड सफलतापूर्वक जोड़ दिया गया है!")

# 2. सभी छात्रों के रिकॉर्ड दिखाने का फंक्शन
def display_students():
    print("\n--- All Student Records ---")
    if not student_records:
        print("कोई रिकॉर्ड उपलब्ध नहीं है!")
    else:
        for index, student in enumerate(student_records, start=1):
            print(f"{index}. Name: {student['name']} | Roll No: {student['roll_no']} | Course: {student['course']}")
        print("-----")

# 3. नाम से छात्र को खोजने का फंक्शन
def search_student():
    search_name = input("Enter the name of the student to search: ")
    found = False
    print("\n--- Search Results ---")
    for student in student_records:
        if student["name"].lower() == search_name.lower():
            print(f"✓ Found! Name: {student['name']} | Roll No: {student['roll_no']} | Course: {student['course']}")
            found = True
    if not found:
        print("✗ इस नाम का कोई छात्र नहीं मिला!")
    print("-----")
```

```

# 4. छात्र का रिकॉर्ड हटाने का फंक्शन
def delete_student():
    delete_name = input("Enter the name of the student to delete: ")
    for student in student_records:
        if student["name"].lower() == delete_name.lower():
            student_records.remove(student)
            print(f"🗑️ {delete_name} का रिकॉर्ड सफलतापूर्वक हटा दिया गया है।")
            return
    print("❌ इस नाम का कोई छात्र रिकॉर्ड में नहीं है।")

# मुख्य प्रोग्राम (Main Menu Loop)
while True:
    print("\n=== Student Record Management System ===")
    print("1. Add Student")
    print("2. Display All Students")
    print("3. Search Student")
    print("4. Delete Student")
    print("5. Exit")

    choice = input("Enter your choice (1-5): ")

    if choice == "1":
        add_student()
    elif choice == "2":
        display_students()
    elif choice == "3":
        search_student()
    elif choice == "4":
        delete_student()
    elif choice == "5":
        print("🎉 थैंक यू! आपका फाइनल प्रोजेक्ट सफलतापूर्वक बंद हो गया है।")
        break
    else:
        print("❌ गलत विकल्प! कृपया 1 से 5 तक ही चुनें।")

```

OUTPUT TASK 9

C:\Users\hp\Desktop\.venv\Scripts\python.exe
"C:\Users\hp\Desktop\task 9.py"

=== Student Record Management System ===

1. Add Student

2. Display All Students

3. Search Student

4. Delete Student

5. Exit

Enter your choice (1-5): 1

Enter student name: AYUSH GUPTA

Enter roll number: 0884

Enter course: COMPUTER SCIENCE

✓ AYUSH GUPTA का रिकॉर्ड सफलतापूर्वक जोड़ दिया गया है!

=== Student Record Management System ===

1. Add Student

2. Display All Students

3. Search Student

4. Delete Student

5. Exit

Enter your choice (1-5): 2

--- All Student Records ---

1. Name: AYUSH GUPTA | Roll No: 0884 | Course: COMPUTER
SCIENCE

=== Student Record Management System ===

1. Add Student

2. Display All Students

3. Search Student

4. Delete Student

5. Exit

Enter your choice (1-5): 3

Enter the name of the student to search: AYUSH GUPTA

--- Search Results ---

✓ Found! Name: AYUSH GUPTA | Roll No: 0884 | Course:
COMPUTER SCIENCE

=== Student Record Management System ===

1. Add Student

2. Display All Students

3. Search Student

4. Delete Student

5. Exit

Enter your choice (1-5): 4

Enter the name of the student to delete: AYUSH GUPTA

🗑 AYUSH GUPTA का रिकॉर्ड सफलतापूर्वक हटा दिया गया है।

=== Student Record Management System ===

1. Add Student

2. Display All Students

3. Search Student

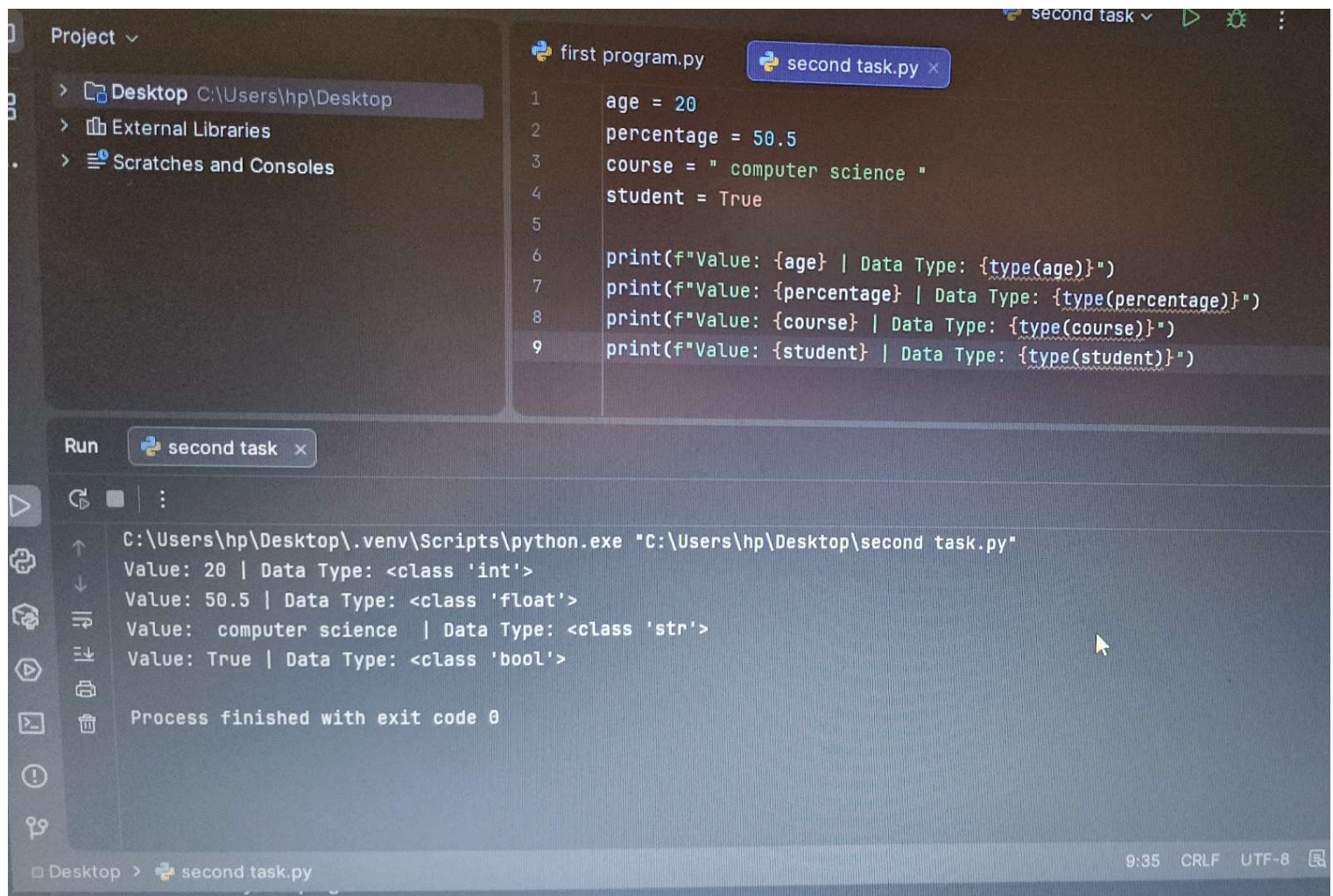
4. Delete Student

5. Exit

Enter your choice (1-5): 5

थैंक यू! आपका फाइनल प्रोजेक्ट सफलतापूर्वक बंद हो गया है।

Process finished with exit code 0



Run

first program x



C:\Users\hp\Desktop\.venv\Scripts\python.exe "C:\Users\hp\Desktop\first program.py"

welcome to my first python program

enter your name: *AYUSH GUPTA*

enter your college name: *THAKUR SHIVKUMAR ENGINEERING COLLEGE BURHANPUR MP*

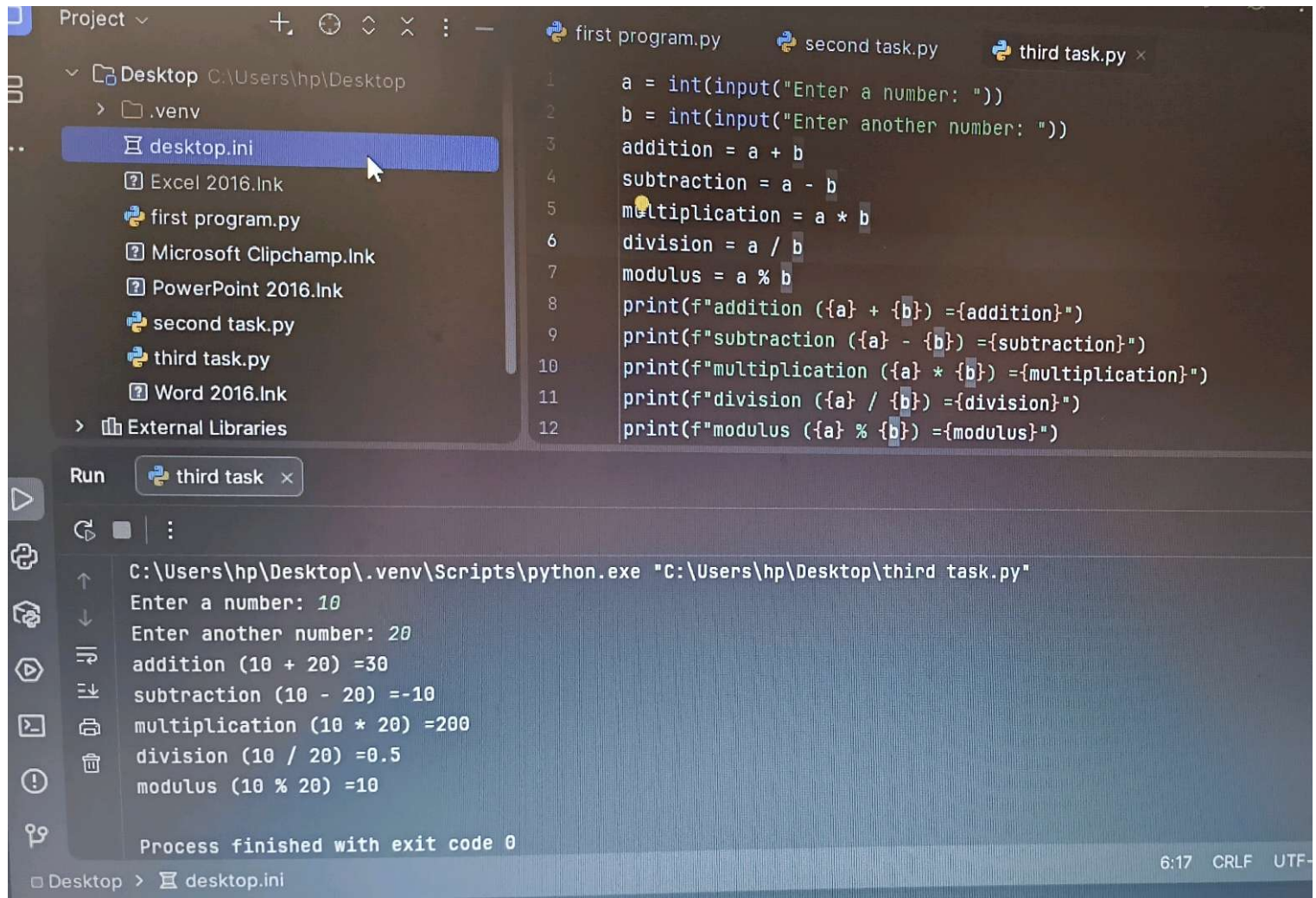
enter your branch name: *COMPUTER SCIENCE*

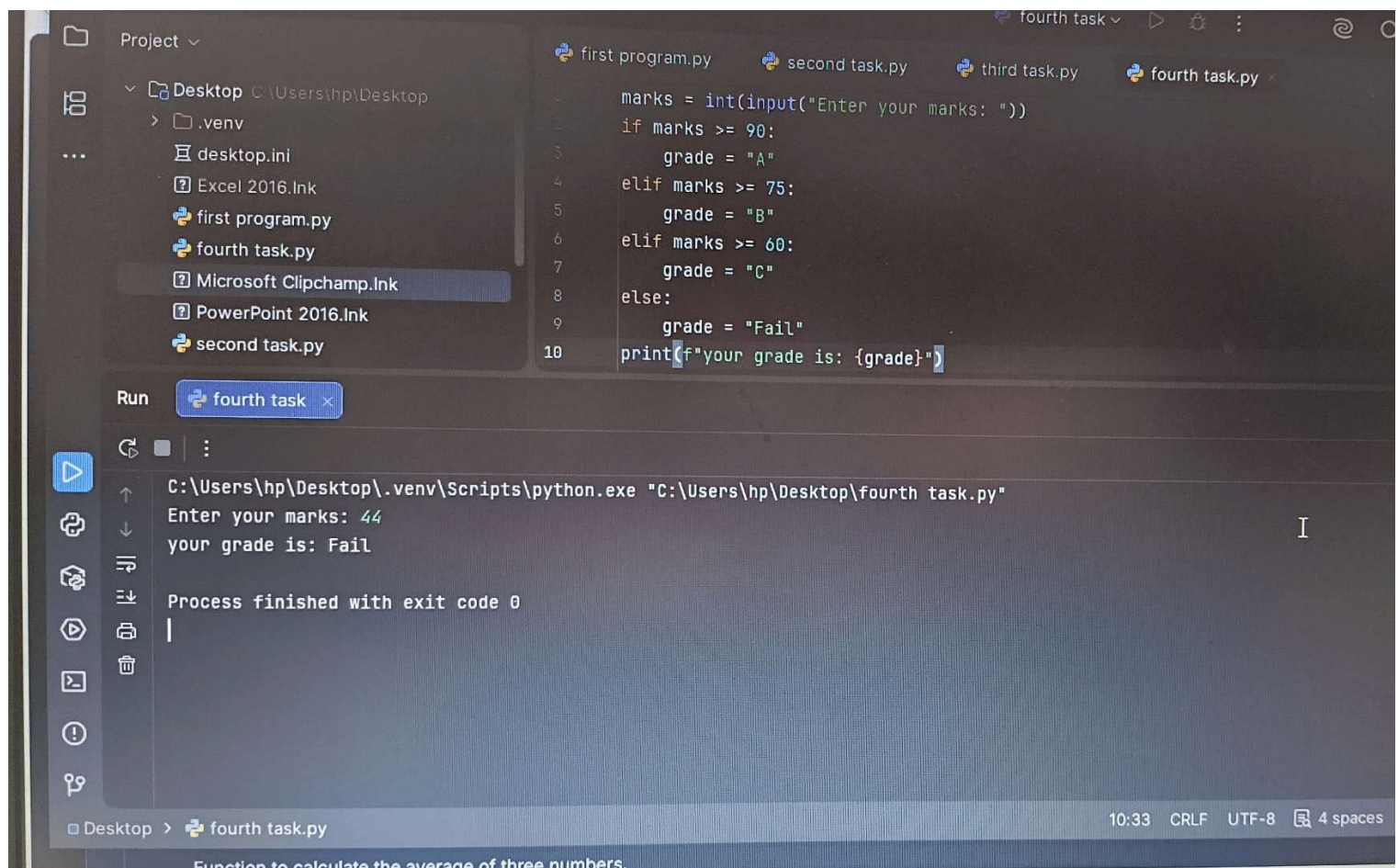
Name: AYUSH GUPTA

college_name: THAKUR SHIVKUMAR ENGINEERING COLLEGE BURHANPUR MP

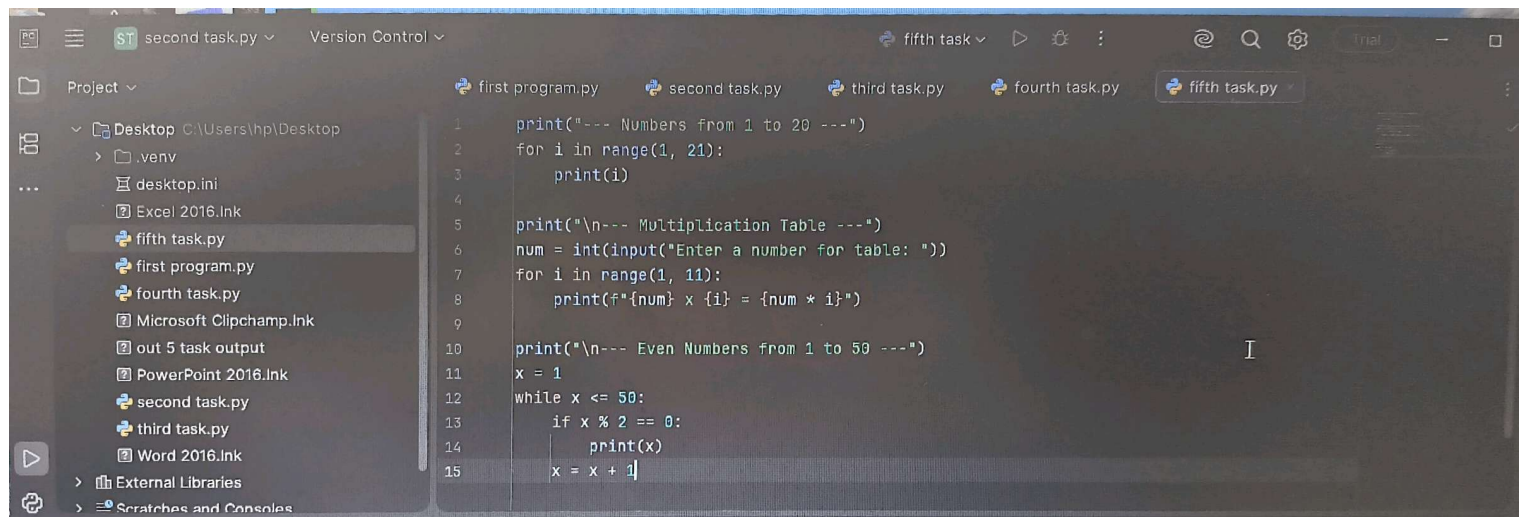
branch_name: COMPUTER SCIENCE

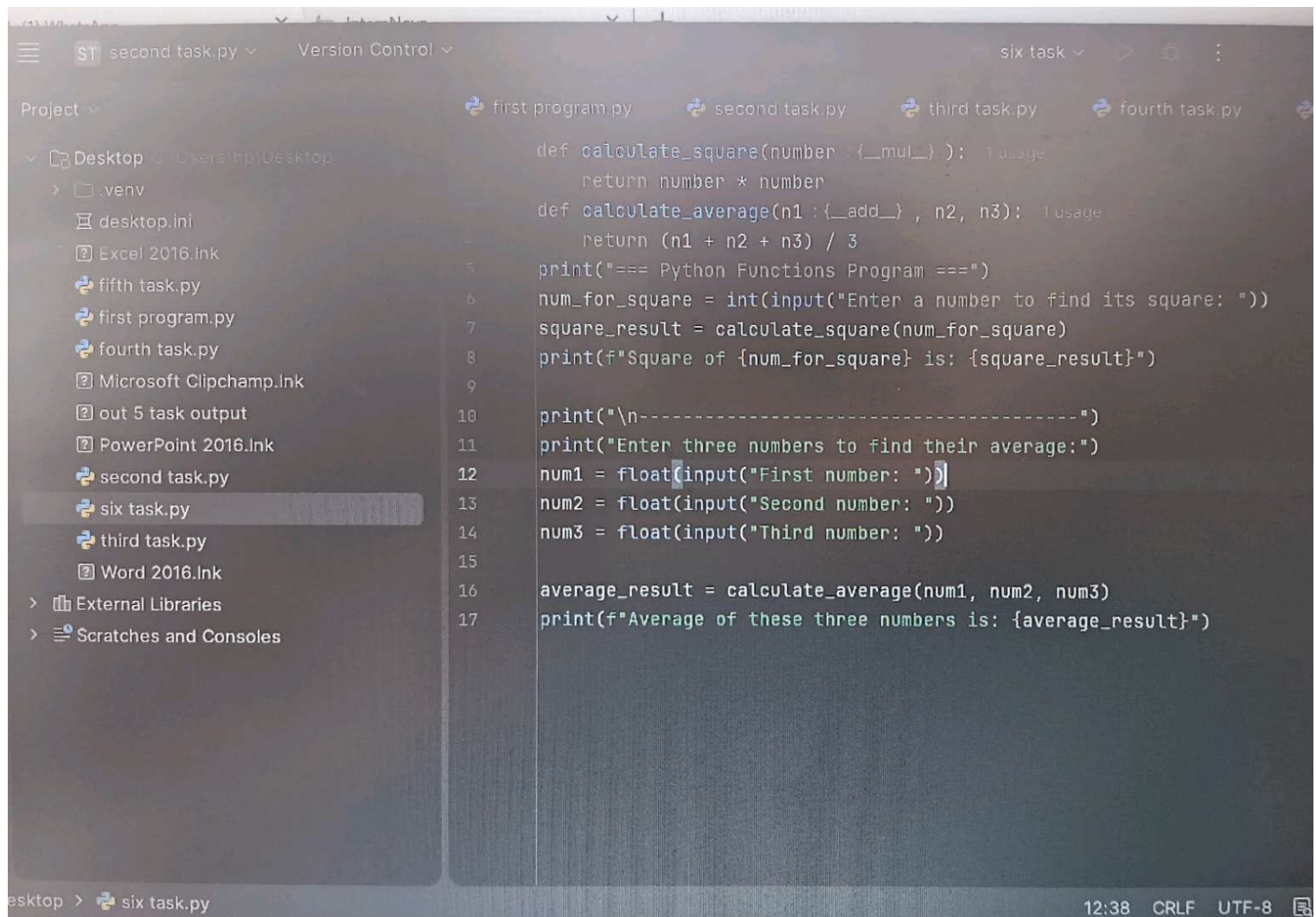
Process finished with exit code 0





Function to calculate the average of three numbers.





```
C:\Users\hp\Desktop\.venv\Scripts\python.exe "C:\Users\hp\Desktop\six task.py"
=== Python Functions Program ===
Enter a number to find its square: 64
Square of 64 is: 4096

-----
Enter three numbers to find their average:
First number: 3
Second number: 6
Third number: 9
Average of these three numbers is: 6.0

Process finished with exit code 0
```

The image shows a Python IDE interface with a project explorer on the left, a code editor in the center, and a run console at the bottom.

Project Explorer: The left sidebar shows a project named 'Desktop' located at 'C:\Users\hp\Desktop'. It contains several files and folders, including '.venv', 'desktop.ini', 'Excel 2016.lnk', 'fifth task.py', 'first program.py', 'fourth task.py', 'Microsoft Clipchamp.lnk', 'my_intro.txt', 'out seven', and 'out 5 task output'.

Code Editor: The main area displays the code for 'task 8.py'. The code is as follows:

```
1 print("=== File Handling Program ===")
2
3 with open("my_intro.txt", "w") as file:
4     file.write("Hello, my name is Ayush Gupta.\n")
5     file.write("I am studying Computer Science Engineering.\n")
6     file.write("This is a text file created using Python.")
7
8 print("\n--- Reading File Contents ---")
9
10 with open("my_intro.txt", "r") as file:
11     content = file.read()
12     print(content)
```

Run Console: The bottom panel shows the execution of the program. The command executed is 'C:\Users\hp\Desktop\.venv\Scripts\python.exe "C:\Users\hp\Desktop\task 8.py"'. The output is:

```
=== File Handling Program ===

--- Reading File Contents ---
Hello, my name is Ayush Gupta.
I am studying Computer Science Engineering.
This is a text file created using Python.

Process finished with exit code 0
```

Run

task 8 x



```
C:\Users\hp\Desktop\.venv\Scripts\python.exe "C:\Users\hp\Desktop\task 8.py"
```

```
=== File Handling Program ===
```

```
--- Reading File Contents ---
```

```
Hello, my name is Ayush Gupta.
```

```
I am studying Computer Science Engineering.
```

```
This is a text file created using Python.
```

```
Process finished with exit code 0
```

sktop > task 8.py

📌 असाइनमेंट विवरण (Brief Explanation) —

Task 1

मैं हमें एक वेलकम मैसेज और Name, College name and branch Name इनपुट लेना था। जिसमें मैंने input() function और f-string का use किया। f का use इसलिए किया क्योंकि f लगाते हैं, अगर नहीं लगाते तो उसे Computer नहीं understand करता, इसलिए f-string का use किया।

Task 2

डेटा टाइप्स की पहचान के लिए type() फंक्शन लगाया। जिसमें integer, float, string और Boolean का output निकाला है। type() फंक्शन कंप्यूटर के स्कैनर जैसा है जो बताता है कि अंदर का डेटा किस टाइप का है।

Task 3

मैं int() का use किया और इसमें modulus (%) का उपयोग remainder (शेषफल) निकालने के लिए किया है, जो भाग करने के बाद नीचे बच जाता है।

Task 4

मैं ग्रेड निकालने के लिए if-elif-else की चेन लगाई है। इसमें मजेदार बात यह है कि कंप्यूटर को जैसे ही सही ग्रेड मिल जाती है, वह बाकी की लाइनें छोड़ देता है और सीधे नीचे कूद जाता है ताकि रिजल्ट एकदम सही आए।

Task 5

मैं लूप का use किया। for लूप से 1 से 20 तक गिनती प्रिंट की और एक पहाड़ा (Table) बनाया। फिर while लूप से 1 से 50 तक के केवल Even (सम) नंबर का आउटपुट निकाला।

Task 6

मैं फंक्शन बनाया। def लिखकर स्क्वायर और एवरेज निकालने की दो मशीनें (Functions) तैयार कीं। इसमें return का use किया जो कैलकुलेशन के बाद असली जवाब को बाहर भेजता है।

Task 7

मैं Strings और Collections को समझा। इसमें List [] (सामान जोड़ने-हटाने के लिए), Tuple () (बदला न जाने वाला बॉक्स), Dictionary (नाम और जानकारी का जोड़ा) और Set {} (बिना डुप्लीकेट वाला) का use करके दिखाया।

Task 8

में File Handling किया। पाइथन कोड से कंप्यूटर के अंदर एक असली टेक्स्ट फाइल (my_intro.txt) बनाई, उसमें अपना इंट्रोडक्शन लिखा (w mode) और फिर उसे वापस रीड (r mode) करके स्क्रीन पर प्रिंट कराया।

Task 9

में एक मिनी प्रोजेक्ट बनाया जिसका नाम Student Record Management System है। इसमें ऊपर सीखे गए सभी टूल्स को मिलाकर छात्र का नाम जोड़ने (Add), लिस्ट देखने (Display), खोजने (Search) और हटाने (Delete) का पूरा सिस्टम तैयार किया है।

निष्कर्ष और आभार:

इस पूरे असाइनमेंट को पूरा करने में मैंने अपनी मेहनत के साथ-साथ AI (Artificial Intelligence) और YouTube की भी मदद ली है। शुरुआत के कुछ बुनियादी टास्क मैंने अपनी समझ और मेहनत से खुद कोड किए, और बाकी के थोड़े एडवांस व पेचीदा टास्क को सही तरीके से समझने, उनके लॉजिक्स को क्लियर करने और एरर्स (Errors) को ठीक करने के लिए मैंने AI टूल्स और YouTube ट्यूटोरियल्स का सहारा लिया। इस मिश्रित प्रयास से मुझे पाइथन प्रोग्रामिंग को गहराई से सीखने और इस प्रोजेक्ट को सफलतापूर्वक पूरा करने में बहुत मदद मिली।

7 or 9 task hard tha isme mene puri madad li youtube or ai ki me iski practice kr rha hu python beginner course bhi start he